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SEMICONDUCTOR DEVICE AND MANUFACTURING METHOD THEREOF

Abstract

A semiconductor device includes a gate structure formed over a channel region of the semiconductor device, a source/drain region adjacent the channel region, and an electrically conductive contact layer over the source/drain region. The source/drain region includes a first epitaxial layer having a first material composition and a second epitaxial layer formed over the first epitaxial layer. The second epitaxial layer has a second material composition different from the first composition. The electrically conductive contact layer is in contact with the first and second epitaxial layers. A bottom of the electrically conductive contact layer is located below an uppermost portion of the first epitaxial layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of U.S. patent application Ser. No. 17/723,033 filed on Apr. 18, 2022, which is a continuation application of U.S. patent application Ser. No. 16/417,051 filed on May 20, 2019, now U.S. Pat. No. 11,309,418, which is a divisional application of U.S. patent application Ser. No. 15/725,091 filed on Oct. 4, 2017, now U.S. Pat. No. 10,297,690, which claims priority to U.S. Provisional Patent Application 62/441,028 filed Dec. 30, 2016, the entire disclosure of each of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] The disclosure relates to a semiconductor integrated circuit, and more particularly to a semiconductor device having an epitaxial source/drain (S/D) structure with voids and its manufacturing process.

BACKGROUND

[0003] As the semiconductor industry has progressed into nanometer technology process nodes in pursuit of higher device density, higher performance, and lower costs, challenges from both fabrication and design issues have resulted in the development of three-dimensional designs, such as a multi-gate field effect transistor (FET) including a FinFET (FinFET). In a FinFET, a gate electrode is adjacent to three side surfaces of a channel region with a gate dielectric layer interposed therebetween. As the size of the FinFET shrinks, the electrode contact area on the S/D shrinks, thereby increasing the contact resistance. As transistor dimensions are continually scaled down, further improvements of the FinFET are required.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The present disclosure is best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale and are used for illustration purposes only. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0005] FIG. 1 shows one of various stages for manufacturing a semiconductor FinFET device according to embodiments of the present disclosure.

[0006] FIG. 2 shows one of various stages for manufacturing a semiconductor FinFET device according to embodiments of the present disclosure.

[0007] FIG. 3 shows one of various stages for manufacturing a semiconductor FinFET device according to embodiments of the present disclosure.

[0008] FIG. 4 shows one of various stages for manufacturing a semiconductor FinFET device according to embodiments of the present disclosure.

[0009] FIGS. 5A-5C show one of various stages for manufacturing a semiconductor FinFET device according to embodiments of the present disclosure. FIG. 5A is a perspective view, FIG. 5B is a cross sectional view along line a-a of FIG. 5A and FIG. 5C is a cross sectional view along line b-b of FIG. 5A.

[0010] FIG. 6 shows one of various stages for manufacturing a semiconductor FinFET device according to embodiments of the present disclosure.

[0011] FIGS. 7A, 7B and 7C show one of various stages for manufacturing a semiconductor FinFET device according to embodiments of the present disclosure.

[0012] FIG. 8 shows one of various stages for manufacturing a semiconductor FinFET device according to embodiments of the present disclosure.

[0013] FIG. 9 shows one of various stages for manufacturing a semiconductor FinFET device according to embodiments of the present disclosure.

[0014] FIG. 10 shows one of various stages for manufacturing a semiconductor FinFET device according to embodiments of the present disclosure.

[0015] FIG. 11 shows one of various stages for manufacturing a semiconductor FinFET device according to embodiments of the present disclosure.

[0016] FIG. 12 shows one of various stages for manufacturing a semiconductor FinFET device according to embodiments of the present disclosure.

[0017] FIG. 13 shows one of various stages for manufacturing a semiconductor FinFET device according to embodiments of the present disclosure.

[0018] FIG. 14 shows one of various stages for manufacturing a semiconductor FinFET device according to embodiments of the present disclosure.

[0019] FIGS. 15A and 15B show cross sectional views showing source/drain structures.

[0020] FIG. 16 shows experimental results of source/drain contact resistance.

DETAILED DESCRIPTION

[0021] It is to be understood that the following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific embodiments or examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, dimensions of elements are not limited to the disclosed range or values, but may depend upon process conditions and/or desired properties of the device. Moreover, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed interposing the first and second features, such that the first and second features may not be in direct contact. Various features may be arbitrarily drawn in different scales for simplicity and clarity. In the accompanied drawings, some layers/features may be omitted for simplification.

[0022] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The device may be otherwise oriented (rotated 90 degrees or at

other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. In addition, the term “made of” may mean either “comprising” or “consisting of.” Further, in the following fabrication process, there may be one or more additional operations in/between the described operations, and the order of operations may be changed.

[0023] FIGS. **1-14** show views of various stages for manufacturing a semiconductor FinFET device according to embodiments of the present disclosure. It is understood that additional operations can be provided before, during, and after processes shown by FIGS. **1-14**, and some of the operations described below can be replaced or eliminated, for additional embodiments of the method. The order of the operations/processes may be interchangeable.

[0024] In FIGS. **1-3**, one or more fin structures are formed. A mask layer **15** is formed over a substrate **10**. The mask layer **15** is formed by, for example, a thermal oxidation process and/or a chemical vapor deposition (CVD) process. The substrate **10** is, for example, a p-type silicon or germanium substrate with an impurity concentration in a range from about 1×10^{15} atoms/cm³ to about 1×10^{16} atoms/cm³. In other embodiments, the substrate is an n-type silicon or germanium substrate with an impurity concentration in a range from about 1×10^{15} atoms/cm³ to about 1×10^{16} atoms/cm³.

[0025] Alternatively, the substrate **10** may comprise another elementary semiconductor, such as germanium; a compound semiconductor including Group IV-IV compound semiconductors such as SiC and SiGe, Group III-V compound semiconductors such as GaAs, GaP, GaN, InP, InAs, InSb, GaAsP, AlGaN, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof. In one embodiment, the substrate **10** is a silicon layer of an SOI (silicon-on insulator) substrate.

Amorphous substrates, such as amorphous Si or amorphous SiC, or insulating material, such as silicon oxide may also be used as the substrate **10**. The substrate **10** may include various regions that have been suitably doped with impurities (e.g., p-type or n-type conductivity).

[0026] The mask layer **15** includes, for example, a pad oxide (e.g., silicon oxide) layer **15A** and a silicon nitride mask layer **15B** in some embodiments.

[0027] The pad oxide layer **15A** may be formed by using thermal oxidation or a CVD process. The silicon nitride mask layer **15B** may be formed by a physical vapor deposition (PVD), such as a sputtering method, a CVD, plasma-enhanced chemical vapor deposition (PECVD), an atmospheric pressure chemical vapor deposition (APCVD), a low-pressure CVD (LPCVD), a high density plasma CVD (HDPCVD), an atomic layer deposition (ALD), and/or other processes.

[0028] The thickness of the pad oxide layer **15A** is in a range from about 2 nm to about 15 nm and the thickness of the silicon nitride mask layer **15B** is in a range from about 2 nm to about 50 nm in some embodiments. A mask pattern is further formed over the mask layer. The mask pattern is, for example, a resist pattern formed by lithography operations.

[0029] By using the mask pattern as an etching mask, a hard mask pattern **15** of the pad oxide layer and the silicon nitride mask layer is formed, as shown in FIG. **1**.

[0030] Then, as shown in FIG. **2**, by using the hard mask pattern **15** as an etching mask, the substrate **10** is patterned into fin structures **20** by trench etching using a dry etching method and/or a wet etching method.

[0031] In other embodiments, The fin structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes.

[0032] Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers, or mandrels, may then be used to pattern the fin structures.

[0033] In FIG. **2**, three fin structures **20** are disposed over the substrate **10**. However, the number of

the fin structures is not limited to three. The numbers may be as small as one or more than three. In addition, one or more dummy fin structures may be disposed adjacent both sides of the fin structure **20** to improve pattern fidelity in patterning processes.

[0034] The fin structure **20** may be made of the same material as the substrate **10** and may continuously extend from the substrate **10**. In this embodiment, the fin structure is made of Si. The silicon layer of the fin structure **20** may be intrinsic, or appropriately doped with an n-type impurity or a p-type impurity.

[0035] The width **W1** of the fin structure **20** is in a range from about 5 nm to about 40 nm in some embodiments, and is in a range from about 7 nm to about 12 nm in other embodiments. The space **S1** between two fin structures is in a range from about 10 nm to about 50 nm in some embodiments. The height (along the Z direction) of the fin structure **20** is in a range from about 100 nm to about 300 nm in some embodiments, and is in a range from about 50 nm to 100 nm in other embodiments.

[0036] The lower part of the fin structure **20** under the gate structure **40** (see, FIG. 5A) may be referred to as a well region, and the upper part of the fin structure **20** may be referred to as a channel region. Under the gate structure **40**, the well region is embedded in the isolation insulating layer **30** (see, FIG. 5A), and the channel region protrudes from the isolation insulating layer **30**. A lower part of the channel region may also be embedded in the isolation insulating layer **30** to a depth of about 1 nm to about 5 nm.

[0037] The height of the well region is in a range from about 60 nm to 100 nm in some embodiments, and the height of the channel region is in a range from about 40 nm to 60 nm, and is in a range from about 38 nm to about 55 nm in other embodiments.

[0038] After the fin structures **20** are formed, the substrate **10** is further etched to form a mesa shape **10M** in some embodiments, as shown in FIG. 3. In other embodiments, the mesa shape **10M** is first formed, and then the fin structures **20** are formed. In certain embodiments, the mesa shape is not formed and the fin structures **20** directly protrude from the substrate **10**.

[0039] After the fin structures **20** and the mesa shape **10M** are formed, the isolation insulating layer **30** is formed in spaces between the fin structures and/or a space between one fin structure and another element formed over the substrate **10**. The isolation insulating layer **30** may also be called a “shallow-trench-isolation (STI)” layer. The insulating material for the isolation insulating layer **30** may include one or more layers of silicon oxide, silicon nitride, silicon oxynitride (SiON), SiOCN, fluorine-doped silicate glass (FSG), or a low-k dielectric material. The isolation insulating layer is formed by LPCVD (low pressure chemical vapor deposition), plasma-CVD or flowable CVD. In the flowable CVD, flowable dielectric materials instead of silicon oxide may be deposited.

Flowable dielectric materials, as their name suggest, can “flow” during deposition to fill gaps or spaces with a high aspect ratio. Usually, various chemistries are added to silicon-containing precursors to allow the deposited film to flow. In some embodiments, nitrogen hydride bonds are added. Examples of flowable dielectric precursors, particularly flowable silicon oxide precursors, include a silicate, a siloxane, a methyl silsesquioxane (MSQ), a hydrogen silsesquioxane (HSQ), an MSQ/HSQ, a perhydrosilazane (TCPS), a perhydro-polysilazane (PSZ), a tetraethyl orthosilicate (TEOS), or a silyl-amine, such as trisilylamine (TSA). These flowable silicon oxide materials are formed in a multiple-operation process. After the flowable film is deposited, it is cured and then annealed to remove un-desired element(s) to form silicon oxide. When the un-desired element(s) is removed, the flowable film densifies and shrinks. In some embodiments, multiple anneal processes are conducted. The flowable film is cured and annealed more than once. The flowable film may be doped with boron and/or phosphorous.

[0040] The insulating layer **30** is first formed in a thick layer so that the fin structures are embedded in the thick layer, and the thick layer is recessed so as to expose the upper portions of the fin structures **20** in some embodiments, as shown in FIG. 4. The height **H1** of the fin structures from the upper surface of the isolation insulating layer **30** is in a range from about 20 nm to about

100 nm in some embodiments, and is in a range from about 30 nm to about 50 nm in other embodiments. After or before recessing the isolation insulating layer **30**, a thermal process, for example, an anneal process, may be performed to improve the quality of the isolation insulating layer **30**. In certain embodiments, the thermal process is performed by using rapid thermal annealing (RTA) at a temperature in a range from about 900° C. to about 1050° C. for about 1.5 seconds to about 10 seconds in an inert gas ambient, such as an N.sub.2, Ar or He ambient.

[0041] After the insulating layer **30** is formed, a gate structure **40** is formed over the fin structures **20**, as shown in FIGS. 5A-5C. FIG. 5A is an exemplary perspective view, FIG. 5B is an exemplary cross sectional view along line a-a of FIG. 5A and FIG. 5C is an exemplary cross sectional view along line b-b of FIG. 5A. FIGS. 6-14 are also exemplary cross sectional views along line b-b of FIG. 5A.

[0042] As shown in FIG. 5A, the gate structure **40** extends in the X direction, while the fin structures **20** extend in the Y direction.

[0043] To fabricate the gate structure **40**, a dielectric layer and a poly silicon layer are formed over the isolation insulating layer **30** and the exposed fin structures **20**, and then patterning operations are performed so as to obtain gate structures including a gate pattern **44** made of poly silicon and a dielectric layer **42**. In some embodiments, the polysilicon layer is patterned by using a hard mask and the hard mask remains on the gate pattern **44** as a cap insulating layer **46**. The hard mask (cap insulating layer **46**) includes one or more layers of insulating material. The cap insulating layer **46** includes a silicon nitride layer formed over a silicon oxide layer in some embodiments. In other embodiments, the cap insulating layer **46** includes a silicon oxide layer formed over a silicon nitride layer. The insulating material for the cap insulating layer **46** may be formed by CVD, PVD, ALD, e-beam evaporation, or other suitable process. In some embodiments, the dielectric layer **42** may include one or more layers of silicon oxide, silicon nitride, silicon oxy-nitride, or high-k dielectrics. In some embodiments, a thickness of the dielectric layer **42** is in a range from about 2 nm to about 20 nm, and in a range from about 2 nm to about 10 nm in other embodiments. The height H2 of the gate structure is in a range from about 50 nm to about 400 nm in some embodiments, and is in a range from about 100 nm to 200 nm in other embodiments.

[0044] In some embodiments, a gate replacement technology is employed. In such a case, the gate pattern **44** and the dielectric layer **42** are a dummy gate electrode and a dummy gate dielectric layer, respectively, which are subsequently removed. If a gate-first technology is employed, the gate pattern **44** and the dielectric layer **42** are used as a gate electrode and a gate dielectric layer.

[0045] Further, gate sidewall spacers **48** are formed on both sidewalls of the gate pattern **44**. The sidewall spacers **48** include one or more layers of insulating material, such as SiO.sub.2, SiN, SiON, SiOCN or SiCN, which are formed by CVD, PVD, ALD, e-beam evaporation, or other suitable process. A low-k dielectric material may be used as the sidewall spacers. The sidewall spacers **48** are formed by forming a blanket layer of insulating material and performing anisotropic etching. In one embodiment, the sidewall spacer layers are made of silicon nitride based material, such as SiN, SiON, SiOCN or SiCN.

[0046] Then, as shown in FIG. 6, a spacer material layer **50** is formed over the fin structures **20**. The spacer material layer **50** is made of dielectric material including silicon nitride based material, such as SiN, SiON, SiOCN or SiOC. In one embodiment, SiN is used as the spacer material layer **50**. The spacer material layer **50** is formed by CVD, PVD, ALD, e-beam evaporation, or other suitable process. The thickness of the spacer material layer **50** is in a range from about 30 nm to about 70 nm in some embodiments.

[0047] In some embodiments, the spacer material layer **50** and the sidewall spacers **48** for the gate structure are separately formed. In other embodiments, the same blanket layer is used for the spacer material layer **50** and the sidewall spacers **48**.

[0048] After forming the spacer material layer **50**, the upper portion of the fin structures **20** are recessed and a part of the spacer material layer **50** disposed on side surfaces and the top surface of

the fin structures protruding from the isolation insulating layer are removed by a dry etching and/or a wet etching operation. The upper portion of the fin structures **20** are recessed (etched) down to the level equal to or below the upper surface of the spacer material layer **50** on the upper surface isolation insulating layer **30**, as shown in FIG. 7A. By adjusting etching conditions, for example, an over-etching time, the spacer material layer **50** remains on the upper surface of the isolation insulating layer **30**. The thickness of the remaining spacer material layer **50** is in a range from about 2 nm to about 10 nm in some embodiments.

[0049] In some embodiments, lower portions of the spacer material layer **50** disposed on sidewall of the fin structures **20** remain without etched away thereby forming fin spacers **55** having a sleeve-like shape, as shown in FIG. 7A. The height H3 of the fin spacers **55** is in a range from about 1 nm to about 10 nm in some embodiments, and is in a range from about 5 nm to about 40 nm in some embodiments.

[0050] In some embodiments, the spacer material layers **50** disposed over the upper surface of the isolation insulating layer **30** is fully removed, but the fin spacers **55** remain, as shown in FIG. 7B. In other embodiments, the spacer material layer **50** is fully removed, as shown in FIG. 7C. In certain embodiments, the spacer material layer **50** remains only on the upper surface of the isolation insulating layer **30** and no sleeve-like portion exists.

[0051] In the following embodiments, the subsequent manufacturing operations are performed on the structure shown in FIG. 7A. However, the same or substantially similar operations can be performed on the structures shown in FIGS. 7B and 7C.

[0052] As shown in FIG. 8, first epitaxial source/drain layers **60** are formed over the recessed fin structures **20**, respectively. The first epitaxial source/drain layers **60** are made of one or more layers of semiconductor material having a different lattice constant than the fin structures **20** (channel regions). When the fin structures are made of Si, the first epitaxial source/drain layers **60** include SiP, SiC or SiCP for an n-channel FinFET and SiGe or Ge for a p-channel FinFET. The first epitaxial source/drain layers **60** are epitaxially formed over the upper portions of the recessed fin structures.

[0053] As shown in FIG. 8, the first source/drain epitaxial layers **60** have a bar shape in the cross section along the X direction, and are separated from each other. The aspect ratio H4/W2 of the first source/drain epitaxial layers is in a range from about 2 to about 15 in some embodiments, where H4 is a height of the first source/drain epitaxial layers **60** from the upper surface of the isolation insulating layer **30** and W2 is a largest width of the first epitaxial layer along the X direction.

[0054] The bar shaped epitaxial layer **60** can be formed by using a high-temperature deposition technique. The high temperature deposition process can promote the growth of the crystallographic facet (100) and possibly impede the growth of the crystallographic facets (110) and (111). The high temperature deposition process can be different for NMOS and PMOS devices. In some embodiments, for NMOS devices, the source/drain epitaxial layer **60** is formed by using a high temperature and pressure deposition process such as a chemical vapor deposition (CVD) process, although other deposition methods can be used in other embodiments. The CVD process can be performed at a high temperature that is within a range of about 650-700° C. and a high pressure that is within a range of about 200-350 Torr. For PMOS devices, the CVD process can be performed at a high temperature that is within a range of about 600-650° C. and a pressure within a range of about 5-50 Torr.

[0055] After the first source/drain epitaxial layers **60** are formed, a second source/drain epitaxial layer **62** is formed over the first source/drain epitaxial layers **60**, as shown in FIG. 9. Due to the crystal orientation of the substrate formed into the fin structures **20** and by adjusting epitaxial growth condition, the second epitaxial source/drain layer **62** grows laterally and has a diamond-like shape.

[0056] Due to the relatively small space between the fin structures, the adjacent epitaxial

source/drain structures formed over each of the first epitaxial layers **60** are merged, thereby forming one second source/drain epitaxial layer. Further, a void or a gap (an air gap) **65** is formed by the second source/drain epitaxial layer **62** and the upper surface of the isolation insulating layer **30**, (or and the spacer material layer **50**), as shown in FIG. **9**.

[0057] In particular, due to the spacer material layer **50** on the upper surface of the isolation insulating layer **30**, the height H5 of the void **65** is larger than the case where no spacer material layer **50** remains on the upper surface of the isolation insulating layer **30**. In some embodiments, the height H5 of the void is in a range from about 5 nm to about 40 nm measured from the upper surface of the spacer material layer **50**, and in a range from about 15 nm to about 25 nm in other embodiments. In addition, due to the remaining spacer material layer **50**, the isolation insulating layer **30** is protected during the fin etching.

[0058] Further, in some embodiments, a capping layer **64** is formed on the second source/drain epitaxial layer **62**, as shown in FIG. **10**. In certain embodiments, the capping layer **64** is also a semiconductor epitaxial layer (i.e., a third source/drain epitaxial layer). The thickness of the capping layer is in a range from about 3 nm to about 20 nm in some embodiments.

[0059] In some embodiments, the FinFET is a p-type FET. The first source/drain epitaxial layer **60** includes Si.sub.1-xGe.sub.x and the second source/drain epitaxial layer **62** includes Si.sub.1-yGe.sub.y, where $x < y$. In other words, a Ge concentration is smaller in the first source/drain epitaxial layer **60** than in the second source/drain epitaxial layer **62**. In certain embodiments, the first source/drain epitaxial layer **60** is made of Si.sub.1-xGe.sub.x, where about $0.01 < x < \text{about } 0.4$. In some embodiments, the first source/drain epitaxial layer **60** further includes at least one of B and Ga in an amount of about 5.0×10^{19} atoms/cm³ to about 1.0×10^{21} atoms/cm³.

[0060] In certain embodiments, the second source/drain epitaxial layer **62** includes Si.sub.1-yGe.sub.y, where about $0.2 < y < \text{about } 0.8$. In some embodiments, the second source/drain epitaxial layer **62** further includes at least one of B and Ga in an amount of about 1.0×10^{20} atoms/cm³ to about 3.0×10^{21} atoms/cm³. The amount of B and/or Ga in the first source/drain epitaxial layer **60** is smaller than the amount of B and/or Ga in the second source/drain epitaxial layer **62**.

[0061] In certain embodiments, the capping layer **64** is an epitaxial layer made of Si.sub.1-zGe.sub.z, where $0 \leq z < \text{about } 0.4$, where $z < y$. In other words, a Ge concentration is smaller in the capping layer **64** than in the second source/drain epitaxial layer **62**. In some embodiments, the capping layer **64** further includes at least one of B and Ga in an amount of about 5.0×10^{19} atoms/cm³ to about 1.0×10^{21} atoms/cm³.

[0062] In some embodiments, the FinFET is an n-type FET. In certain embodiments, the first source/drain epitaxial layer **60** comprises Si.sub.1-xC.sub.x and the second epitaxial layer comprises Si.sub.1-yC.sub.y, where $0 < x < y < \text{about } 0.5$. In some embodiments, the first source/drain epitaxial layer **60** and the second source/drain epitaxial layer **62** further include at least one of P and As. An amount of P and/or As in the first source/drain epitaxial layer **60** is smaller than an amount of P and/or As in the second source/drain epitaxial layer **62**.

[0063] In certain embodiments, the first source/drain epitaxial layer **60** includes P and/or As in an amount of about 5.0×10^{19} atoms/cm³ to about 2.0×10^{21} atoms/cm³, and the second source/drain epitaxial layer **62** includes P and/or As in an amount of about 1.0×10^{20} atoms/cm³ to about 8.0×10^{21} atoms/cm³.

[0064] In certain embodiments, the capping layer **64** is an epitaxial layer including Si.sub.1-zC.sub.z, where $0 < z < \text{about } 0.5$. In some embodiments, the capping layer **64** further includes P and/or As in an amount of about 0 to about 1.0×10^{21} atoms/cm³.

[0065] The first to third source/drain epitaxial layers may be grown at a temperature of about 300 to 900° C. under a pressure of about 1 mTorr to 760 Torr or about 1×10^{-7} Torr to about 1×10^{-11} Torr by using a Si containing gas, such as SiH.sub.4, Si.sub.2H.sub.6, Si.sub.3H.sub.8 or SiCl.sub.2H.sub.2; a Ge containing gas, such as GeH.sub.4, Ge.sub.2H.sub.6 or

GeCl.sub.2H.sub.2; a C containing gas, such as CH.sub.4 or C.sub.2H.sub.6; a dopant gas, such as B.sub.2H.sub.6, PH.sub.3 or AsH.sub.3; or an organic metal material, such as trimethylgallium. The source/drain structure for an n-channel FET and the source/drain structure for a p-channel FET may be formed by separate epitaxial processes.

[0066] For the first source/drain epitaxial layer **60**, the epitaxial deposition is performed by enhancing a vertical deposition. In some embodiments, the epitaxial deposition of the first source/drain epitaxial layer **60** is performed using a high temperature and pressure deposition process. The epitaxial process is performed at a high temperature that is within a range of about 500-800° C. and a high pressure that is within a range of about 1 mTorr to about 350 Torr in some embodiments.

[0067] In other embodiments, the first source/drain epitaxial layer **60** can be formed by the combination of epitaxial growth and etching. In this operation, a base epitaxial layer is first formed over the recessed fin structures. The base epitaxial layer grows vertically and laterally, forming a diamond or hexagonal shape, and the side portions are removed, thereby forming a bar shape.

[0068] In some embodiments, the formation process includes an epi-and-etch operation. The epi-and-etch process can be performed the epitaxial deposition at nominal conditions (e.g., to form source/drain epitaxial layer with laterally extended portions) followed by etch operations. After the epitaxial process, lateral portions are removed, thereby forming a bar shape having a top portion with flat side surfaces. In some embodiments, for NMOS device formation, the etch process can be an anisotropic etch operation performed using a mixture of germanium tetrahydride (GeH.sub.4) and hydrochloric acid (HCl) with a GeH.sub.4 to HCl mixture ratio within a range of about 0.5-1.2. The etch operation can be performed at a high temperature within a range of about 650-750° C. and a pressure within a range of about 5-100 Torr. In other embodiments, other etch processes using different process conditions can be used.

[0069] In other embodiments, for PMOS device formation, the etch process can be an etch operation using HCl at a flow rate within a range of about 50-120 sccm. The etch process can be performed at a temperature within a range of about 600-650° C. and a pressure within a range of about 5-50 Torr. In other embodiments, other etch processes using different process conditions can be used.

[0070] After the capping layer **64** is formed, an insulating layer **80**, functioning as a contact etch stop layer (CESL), is formed over the source/drain epitaxial layers, as shown in FIG. **11**. In some embodiments, the CESL **80** is made of, for example, SiN, SiON, SiOCN or SiCN.

[0071] Further, as shown in FIG. **12**, an interlayer dielectric layer **85** is formed over the underlying structure. The ILD layer **85** includes one or more layers of silicon oxide, SiOC, SiOCN or SiCN or other low-k materials, or porous materials, or any other suitable dielectric material.

[0072] Then, by using a patterning operation including lithography and etching, a source/drain contact opening **90** is formed in the interlayer dielectric layer **85**, as shown in FIG. **13**. The etching operations of the ILD layer and the source/drain epitaxial layers are performed by using one or more source gases including HCl, HBr, CF.sub.4, CH.sub.3F, CHF.sub.3, CH.sub.4, SO.sub.2, O.sub.2, He, NF.sub.3 and Cl.sub.2.

[0073] In this patterning operation, the ILD layer **85**, the CESL **80**, the capping layer **64** and the first and second source/drain epitaxial layers **60** and **62** are partially etched. However, since the first source/drain epitaxial layer **60** has a lower Ge (or P or C) concentration, the etching rate of the first source/drain epitaxial layer **60** is smaller than that of the second source/drain epitaxial layer **62**. Accordingly, the first source/drain epitaxial layers **60** protrude from the bottom of the source/drain contact opening **90**, as shown in FIG. **13**. In some embodiments, the protruding amount H6 of the first source/drain epitaxial layers **60** is in a range from about 2 nm to about 20 nm, and is in a range from about 5 nm to about 10 nm in other embodiments.

[0074] Subsequently, a source/drain contact layer **100** made of an electrically conductive material is formed in the source/drain contact opening **90** as shown in FIG. **14**. FIG. **14** also shows an

enlarged view of the circled portion. The source/drain contact layer **100** includes a single layer or multiple layers of any suitable metal such as Co, W, Ti, Ta, Cu, Al and/or Ni and/or nitride thereof. [0075] In some embodiments, a silicide layer **102** is formed over the epitaxial source/drain layers before the contact layer **100** is formed. A metal material, such as Ni, Ti, Ta and/or W, is formed over the source/drain epitaxial layers **60** and **62**, and an annealing operation is performed to form a silicide layer **102**. In other embodiments, a silicide material, such as NiSi, TiSi, TaSi and/or WSi, is formed over the epitaxial source/drain layers, and an annealing operation may be performed. The annealing operation is performed at a temperature of about 250° C. to about 850° C. The metal material or the silicide material is formed by CVD or ALD. The thickness of the silicide layer **102** is in a range from about 4 nm to about 10 nm in some embodiments. In other embodiments, when the contact layer **100** including Ni, Ti, Ta and/or W is formed on the source/drain layers, a silicide layer **102** is simultaneously formed.

[0076] In some embodiments, the conductive source/drain contact layer **100** includes a blanket layer of an adhesive (glue) layer **104** and a body metal layer **106**. The adhesive layer **104** includes one or more layers of conductive materials. In some embodiments, the adhesive layer **104** includes a TiN layer formed on a Ti layer. Any other suitable conductive material can be used. The thickness of each of the TiN layer and the Ti layer is in a range from about 1 nm to about 5 nm in some embodiments. The adhesive layer **104** can be formed by CVD, PVD, ALD, electro-plating, a combination thereof, or other suitable film forming methods. The adhesive layer **104** is used to prevent the body metal layer **106** from peeling off. In some embodiments, no adhesive layer is used and the body metal layer is directly formed in the contact openings. In some embodiments, the Ti layer of the adhesive layer can be used as the metal layer for forming a silicide layer **102**.

[0077] The body metal layer **106** is one of Co, W, Mo and Cu, or any other suitable conductive material in some embodiments. In one embodiment, Co is used as the body metal layer. The body metal layer can be formed by CVD, PVD, ALD, electro-plating, a combination thereof or other suitable film forming methods. After the conductive material layer(s) is formed, a planarization operation, such as chemical mechanical polishing (CMP) or etch-back operations, is performed to remove the excess materials, thereby forming a source/drain contact layer **100**, as shown in FIG. **14**.

[0078] The aspect ratio H7/W3 of the first source/drain epitaxial layers is in a range from about 4 to about 15 in some embodiments, where H7 is a height of the first source/drain epitaxial layers **60** from the upper surface of the isolation insulating layer **30** and W3 is a largest width of the first epitaxial layer along the X direction.

[0079] As shown in FIG. **14**, the first source/drain epitaxial layers **60** protrude from the bottom of the contact opening **90** and penetrate into the contact layer **100**, and the electrically conductive contact layer **100** is in contact with upper surfaces and side surfaces of the upper portions (protruding portions) of the first source/drain epitaxial layers **60**. Further, a bottom of the electrically conductive contact layer **100** is located closer to the substrate than an uppermost portion of the first source/drain epitaxial layers **60**. With these structures, the interfacial area between the contact layer **100** and the source/drain epitaxial layers (or silicide layer) increases, thereby decreasing the contact resistance between the contact layer **100** and the source/drain epitaxial layers (or silicide layer).

[0080] In some embodiments, before the contact opening **90** is formed, a metal gate structure (not shown) is formed. After forming the ILD layer **85**, the dummy gate structure (dummy gate electrode **44** and dummy gate dielectric layer **42**) are removed and replaced with a metal gate structure (metal gate electrode and gate dielectric layer). In certain embodiments, after the ILD layer **85** is formed, the dummy gate electrode **44** and the dummy gate dielectric layer **42** are removed, by appropriate etching processes, respectively, to form a gate opening. Metal gate structures including a gate dielectric layer and metal gate electrode are formed in the gate openings. The gate dielectric layer may be formed over an interface layer (not shown) disposed over the

channel layer of the fin structures **20**. The interface layer may include silicon oxide or germanium oxide with a thickness of 0.2 nm to 1.5 nm in some embodiments. In other embodiments, the thickness of the interface layer is in a range about 0.5 nm to about 1.0 nm. The gate dielectric layer includes one or more layers of dielectric materials, such as silicon oxide, silicon nitride, or high-k dielectric material, other suitable dielectric material, and/or combinations thereof. Examples of high-k dielectric material include HfO_2 , HfSiO , HfSiON , HfTaO , HTaO , HfZrO , zirconium oxide, aluminum oxide, titanium oxide, hafnium dioxide-alumina ($\text{HfO}_2\text{—Al}_2\text{O}_3$) alloy, other suitable high-k dielectric materials, and/or combinations thereof. The gate dielectric layer is formed by, for example, chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), high density plasma CVD (HDPCVD), or other suitable methods, and/or combinations thereof. The thickness of the gate dielectric layer is in a range from about 1 nm to about 10 nm in some embodiments, and may be in a range from about 2 nm to about 7 nm in other embodiments. The metal gate electrode is formed over the gate dielectric layer. The metal gate electrode includes one or more layers of any suitable metal material, such as aluminum, copper, titanium, tantalum, cobalt, molybdenum, tantalum nitride, nickel silicide, cobalt silicide, TiN, WN, TiAl, TiAlN, TaCN, TaC, TaSiN, metal alloys, other suitable materials, and/or combinations thereof. In certain embodiments of the present disclosure, one or more work function adjustment layers (not shown) may be interposed between the gate dielectric layer and the metal gate electrode. The work function adjustment layer is made of a conductive material such as a single layer of TiN, TaN, TaAlC, TiC, TaC, Co, Al, TiAl, HfTi, TiSi, TaSi or TiAlC, or a multilayer of two or more of these materials. For the n-channel FinFET, one or more of TaN, TaAlC, TiN, TiC, Co, TiAl, HfTi, TiSi and TaSi is used as the work function adjustment layer, and for the p-channel FinFET, one or more of TiAlC, Al, TiAl, TaN, TaAlC, TiN, TiC and Co is used as the work function adjustment layer. After depositing appropriate materials for the metal gate structures, planarization operations, such as CMP, are performed.

[0081] After forming the contact layer **100**, further CMOS processes are performed to form various features such as additional interlayer dielectric layers, contacts/vias, interconnect metal layers, and passivation layers, etc.

[0082] In an alternative embodiment, the silicide layer is formed before the contact opening **90** is formed. In such a case, after forming the silicide layer, the ILD layer **85** is formed.

[0083] FIGS. **15A** and **15B** show cross sectional views along the Y direction of the source/drain regions. FIG. **15A** corresponds to the structure of FIG. **14** and FIG. **15B** corresponds to the structure formed without forming the first source/drain epitaxial layer **60**.

[0084] In the structure of FIG. **15B**, since the epitaxial layer **62** contains a relatively higher amount of Ge, metal element(s) of the contact layer **100** may diffuse into the epitaxial layer **62**, causing higher contact resistance. In contrast, in the structure of FIG. **15A**, since the epitaxial layer **60** contains a relatively lower amount of Ge, metal element(s) of the contact layer **100** can be blocked from diffusing into the source/drain epitaxial layers, thereby reducing contact resistance.

[0085] FIG. **16** shows experimental results of source/drain contact resistance between the structures of FIGS. **15A** and **15B**. As shown in FIG. **16**, the contact resistance values of the structure of FIG. **15A** is lower and have smaller variations than the structure of FIG. **15B**. In the foregoing embodiments, the number of fin structures (the number of the first epitaxial layers **60**) per contact layer **100** is three. However, the number of fins per contact layer **100** can be one, two, four, five or more. In multiple fin structures, the second source/drain epitaxial layer **62** is a merged structure with or without a void(s).

[0086] In the present disclosure, since first bar-shaped source/drain epitaxial layers having a lower Ge concentration are formed, it is possible to increase the interfacial area between the source/drain contact and the source/drain epitaxial layer (or silicide layer), thereby decreasing the contact resistance.

[0087] It will be understood that not all advantages have been necessarily discussed herein, no

particular advantage is required for all embodiments or examples, and other embodiments or examples may offer different advantages.

[0088] In accordance with one aspect of the present disclosure, in a method of manufacturing a semiconductor device including a FinFET, a first epitaxial layer is formed at a source/drain region of a fin structure for the FinFET. A second epitaxial layer is formed over the first epitaxial layer. The second epitaxial layer has a material composition different from the first epitaxial layer. An interlayer dielectric (ILD) layer is formed over the first and second epitaxial layers. A contact opening is formed by etching a part of the ILD layer and a part of the second epitaxial layer. The first epitaxial layer protrudes from a bottom of the contact opening. In one or more of the foregoing or following embodiments, an electrically conductive contact layer is further in contact with the first and second epitaxial layers in the contact opening. A bottom of the electrically conductive contact layer is located below an uppermost portion of the first epitaxial layer. In one or more of the foregoing or following embodiments, the FinFET is a p-type, and the first epitaxial layer includes $\text{Si.sub.1-xGe.sub.x}$ and the second epitaxial layer includes $\text{Si.sub.1-yGe.sub.y}$, where $x < y$. In one or more of the foregoing or following embodiments, the first epitaxial layer includes $\text{Si.sub.1-xGe.sub.x}$, where $0.01 < x < 0.4$. In one or more of the foregoing or following embodiments, the first epitaxial layer further includes at least one of B and Ga in an amount of 5.0×10^{19} atoms/cm³ to 1.0×10^{21} atoms/cm³. In one or more of the foregoing or following embodiments, the second epitaxial layer includes $\text{Si.sub.1-yGe.sub.y}$, where $0.2 < y < 0.8$. In one or more of the foregoing or following embodiments, the second epitaxial layer further includes at least one of B and Ga in an amount of 1.0×10^{20} atoms/cm³ to 3.0×10^{21} atoms/cm³. In one or more of the foregoing or following embodiments, the first epitaxial layer and the second epitaxial layer further include at least one of B and Ga, and an amount of at least one of B and Ga in the first epitaxial layer is smaller than an amount of at least one of B and Ga in the second epitaxial layer. In one or more of the foregoing or following embodiments, a capping layer made of a semiconductor material is further formed on the second epitaxial layer. In one or more of the foregoing or following embodiments, the capping layer is an epitaxial layer comprising $\text{Si.sub.1-zGe.sub.z}$, where $0 \leq z < 0.4$, where $z < y$. In one or more of the foregoing or following embodiments, the capping layer further includes at least one of B and Ga in an amount of 5.0×10^{19} atoms/cm³ to 1.0×10^{21} atoms/cm³.

[0089] In accordance with another aspect of the present disclosure, in a method of manufacturing a semiconductor device including a FinFET, a first fin structure and a second fin structure are formed over a substrate such that upper portions of the first and second fin structures protrude from an isolation insulating layer. A gate structure is formed over parts of the first and second fin structures. A spacer material layer is formed on sidewalls of the first and second fin structures protruding from the isolation insulating layer and not covered by the gate structure, and on an upper surface of the isolation insulating layer. Upper portions of the first and second fin structures are recessed, thereby forming a recessed first fin structure and a recessed second fin structure. First epitaxial layers are formed over the recessed first and second fin structures, respectively, the first epitaxial layers being separated from each other. A second epitaxial layer is formed over the first epitaxial layers. The second epitaxial layer has a material composition different from the first epitaxial layer. An interlayer dielectric (ILD) layer is formed over the first and second epitaxial layers. A contact opening is formed by etching a part of the ILD layer and a part of the second epitaxial layer. The first epitaxial layers protrude from a bottom of the contact opening. In one or more of the foregoing or following embodiments, a void is formed between the second epitaxial layer and the upper surface of the isolation insulating layer. In one or more of the foregoing or following embodiments, in the recessing upper portions of the first and second fin structures, lower portions of the spacer material layer disposed on the sidewalls of the first and second fin structures and the spacer material layer disposed on the upper surface of the isolation insulating layer remain. A void is formed between the second epitaxial layer and the remaining spacer material layer on the upper

surface of the isolation insulating layer. In one or more of the foregoing or following embodiments, an electrically conductive contact layer is further formed in contact with the first and second epitaxial layers in the contact opening. A bottom of the electrically conductive contact layer is located below an uppermost portion of the first epitaxial layers. In one or more of the foregoing or following embodiments, the FinFET is a p-type, and the first epitaxial layer includes Si.sub.1-xGe.sub.x and the second epitaxial layer includes Si.sub.1-yGe.sub.y, where $x < y$. In one or more of the foregoing or following embodiments, the first epitaxial layers include Si.sub.1-xGe.sub.x, where $0.01 < x < 0.4$, and the second epitaxial layer includes Si.sub.1-yGe.sub.y, where $0.2 < y < 0.8$. In one or more of the foregoing or following embodiments, the first epitaxial layers further include at least one of B and Ga in an amount of 5.0×10^{19} atoms/cm³ to 1.0×10^{21} atoms/cm³. In one or more of the foregoing or following embodiments, the second epitaxial layer further includes at least one of B and Ga in an amount of 1.0×10^{20} atoms/cm³ to 3.0×10^{21} atoms/cm³. In one or more of the foregoing or following embodiments, an amount of at least one of B and Ga in the first epitaxial layers is smaller than an amount of at least one of B and Ga in the second epitaxial layer. In one or more of the foregoing or following embodiments, an aspect ratio H/W of the first epitaxial layers is in a range from 4 to 20, where H is a height of the first epitaxial layers from the upper surface of the isolation insulating layer and W is a largest width of the first epitaxial layers along a direction in which the gate structure extends.

[0090] In accordance with another aspect of the present disclosure, in a method of manufacturing a semiconductor device including a FinFET, three or more fin structures are formed over a substrate such that upper portions of the fin structures protrude from an isolation insulating layer. A gate structure is formed over parts of the fin structures. Upper portions of the fin structures are recessed, thereby forming recessed fin structures. First epitaxial layers are formed over the recessed fin structures, respectively. The first epitaxial layers are separated from each other. A second epitaxial layer is formed over the first epitaxial layers. The second epitaxial layer has a material composition different from the first epitaxial layer. A third epitaxial layer is formed over the second epitaxial layer. The third epitaxial layer has a material composition different from the second epitaxial layer. An interlayer dielectric (ILD) layer over the first, second and third epitaxial layers. A contact opening is formed by etching a part of the ILD layer and a part of the third and second epitaxial layers. An aspect ratio H/W of the first epitaxial layers is in a range from 4 to 20, where H is a height of the first epitaxial layers from the upper surface of the isolation insulating layer and W is a largest width of the first epitaxial layers along a direction in which the gate structure extends.

[0091] In accordance with one aspect of the present disclosure, a semiconductor device includes a gate structure formed over a channel region of the semiconductor device, a source/drain region adjacent the channel region, and an electrically conductive contact layer over the source/drain region. The source/drain region includes a first epitaxial layer having a first material composition, and a second epitaxial layer formed over the first epitaxial layer, the second epitaxial layer having a second material composition different from the first composition. The electrically conductive contact layer is in contact with the first and second epitaxial layers, and a bottom of the electrically conductive contact layer is located below an uppermost portion of the first epitaxial layer. In one or more of the foregoing or following embodiments, the semiconductor device is a p-type field effect transistor (FET), and the first epitaxial layer includes Si.sub.1-xGe.sub.x and the second epitaxial layer includes Si.sub.1-yGe.sub.y, where $x < y$. In one or more of the foregoing or following embodiments, the first epitaxial layer includes Si.sub.1-xGe.sub.x, where $0.01 < x < 0.4$. In one or more of the foregoing or following embodiments, the first epitaxial layer further includes at least one of B and Ga in an amount of 5.0×10^{19} atoms/cm³ to 1.0×10^{21} atoms/cm³. In one or more of the foregoing or following embodiments, the second epitaxial layer includes Si.sub.1-yGe.sub.y, where $0.2 < y < 0.8$. In one or more of the foregoing or following embodiments, the second epitaxial layer further includes at least one of B and Ga in an amount of 1.0×10^{20} atoms/cm³ to 3.0×10^{21} atoms/cm³. In one or more of the foregoing or following

embodiments, the first epitaxial layer and the second epitaxial layer further include at least one of B and Ga, and an amount of at least one of B and Ga in the first epitaxial layer is smaller than an amount of at least one of B and Ga in the second epitaxial layer. In one or more of the foregoing or following embodiments, a capping layer made of a semiconductor material is further disposed on the second epitaxial layer. In one or more of the foregoing or following embodiments, the capping layer is an epitaxial layer comprising $\text{Si}_{1-z}\text{Ge}_z$, where $0 \leq z < 0.4$, where $z < y$. In one or more of the foregoing or following embodiments, the capping layer further includes at least one of B and Ga in an amount of 5.0×10^{19} atoms/cm³ to 1.0×10^{21} atoms/cm³. In one or more of the foregoing or following embodiments, the semiconductor device is an n-type field effect transistor (FET), and the first epitaxial layer includes $\text{Si}_{1-x}\text{C}_x$ and the second epitaxial layer includes $\text{Si}_{1-y}\text{C}_y$. In one or more of the foregoing or following embodiments, the first epitaxial layer and the second epitaxial layer further include at least one of P and As, and an amount of at least one of P and As in the first epitaxial layer is smaller than an amount of at least one of P and As in the second epitaxial layer. In one or more of the foregoing or following embodiments, the first epitaxial layer includes at least one of P and As in an amount of 5.0×10^{19} atoms/cm³ to 2.0×10^{21} atoms/cm³. In one or more of the foregoing or following embodiments, the second epitaxial layer includes at least one of P and As in an amount of 1.0×10^{20} atoms/cm³ to 8.0×10^{21} atoms/cm³. In one or more of the foregoing or following embodiments, a capping layer made of a semiconductor material is further disposed on the second epitaxial layer. In one or more of the foregoing or following embodiments, the capping layer is an epitaxial layer comprising $\text{Si}_{1-z}\text{C}_z$. In one or more of the foregoing or following embodiments, the capping layer further includes at least one of P and As in an amount of 1.0×10^{20} atoms/cm³ to 8.0×10^{21} atoms/cm³. In one or more of the foregoing or following embodiments, an aspect ratio H/W of the first epitaxial layer is in a range from 4 to 20, where H is a height of the first epitaxial layers from the upper surface of the isolation insulating layer and W is a largest width of the first epitaxial layer along a direction in which the gate structure extends.

[0092] In accordance with another aspect of the present disclosure, a semiconductor device including a FinFET is provided. The FinFET includes: first fin structure and a second fin structure disposed over a substrate, upper portions of the first and second fin structures protruding from an isolation insulating layer; a gate structure over parts of the first and second fin structures; first epitaxial layers disposed over source/drain regions of the first and second fin structures, respectively, the first epitaxial layers being separated from each other; a second epitaxial layer disposed over the first epitaxial layers, the second epitaxial layer having a material composition different from the first epitaxial layer; an interlayer dielectric (ILD) layer disposed over the first and second epitaxial layers; and an electrically conductive contact layer in contact with the first and second epitaxial layers. A bottom of the electrically conductive contact layer is located closer to the substrate than an uppermost portion of the first epitaxial layer. In one or more of the foregoing or following embodiments, a void is formed between the second epitaxial layer and the upper surface of the isolation insulating layer. In one or more of the foregoing or following embodiments, the first epitaxial layers include $\text{Si}_{1-x}\text{Ge}_x$, where $0.01 < x < 0.4$, and the second epitaxial layer includes $\text{Si}_{1-y}\text{Ge}_y$, where $0.2 < y < 0.8$.

[0093] In accordance with another aspect of the present disclosure, a semiconductor device including a FinFET is provided. The FinFET includes: three or more fin structures disposed over a substrate such that upper portions of the fin structures protruding from an isolation insulating layer; a gate structure disposed over parts of the fin structures; first epitaxial layers disposed over source/drain regions of the fin structure, respectively, the first epitaxial layers being separated from each other; a second epitaxial layer disposed over the first epitaxial layers, the second epitaxial layer having a material composition different from the first epitaxial layer; a third epitaxial layer disposed over the second epitaxial layer, the third epitaxial layer having a material composition

different from the second epitaxial layer; and a contact opening disposed in contact with the first, second and third epitaxial layers. The electrically conductive contact layer is in contact with upper surfaces and side surfaces of upper portions of the first epitaxial layers.

[0094] The foregoing outlines features of several embodiments or examples so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments or examples introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor device, comprising: at least one fin structure disposed over a substrate; an isolation insulating layer disposed over the substrate and surrounding the at least one fin structure; a first epitaxial layer having a first material composition disposed over the at least one fin structure; a second epitaxial layer formed over the first epitaxial layer, the second epitaxial layer having a second material composition different from the first composition; and an electrically conductive layer disposed over the first and second epitaxial layers, and the isolation insulating layer, wherein a ratio of a height of the first epitaxial layer along a first direction from an upper surface of the isolation insulating layer to a width of the first epitaxial layer along a second direction crossing the first direction ranges from 2 to 15.
2. The semiconductor device of claim 1, wherein the second epitaxial layer surrounds the first epitaxial layer along the second direction.
3. The semiconductor device of claim 1, wherein the first epitaxial layer penetrates the electrically conductive layer.
4. The semiconductor device of claim 1, further comprising a third epitaxial layer made of a third material composition different from the second material composition and disposed on the second epitaxial layer.
5. The semiconductor device of claim 4, wherein the first epitaxial layer comprises $\text{Si.sub.1-xGe.sub.x}$, the second epitaxial layer comprises $\text{Si.sub.1-yGe.sub.y}$, and the third epitaxial layer comprises $\text{Si.sub.1-zGe.sub.z}$, where $x < y$ and $z < y$.
6. The semiconductor device of claim 5, wherein the third epitaxial layer further comprises B in an amount of 5.0×10^{19} atoms/cm³ to 1.0×10^{21} atoms/cm³.
7. The semiconductor device of claim 5, wherein the third epitaxial layer further comprises Ga in an amount of 5.0×10^{19} atoms/cm³ to 1.0×10^{21} atoms/cm³.
8. The semiconductor device of claim 5, wherein the at least one fin structure is made of Si.
9. A semiconductor device comprising: a mesa structure disposed over a substrate; at least one fin structure disposed over the mesa structure, wherein the at least one fin structure protrudes along a first direction from the mesa structure; an insulating layer surrounding the mesa structure and the at least one fin structure; a first epitaxial layer disposed over at least one fin structure; a second epitaxial layer disposed over the first epitaxial layer, the second epitaxial layer having a material composition different from the first epitaxial layer; a metal-containing contact layer disposed over the first epitaxial layer and first portion of the second epitaxial layer; and a third epitaxial layer disposed over a second portion of the second epitaxial layer, wherein: the first epitaxial layer comprises $\text{Si.sub.1-xGe.sub.x}$, where $0.01 < x < 0.4$, the second epitaxial layer comprises $\text{Si.sub.1-yGe.sub.y}$, where $0.2 < y < 0.8$, the third epitaxial layer comprises $\text{Si.sub.1-zGe.sub.z}$, where $x < y$ and $z < y$, and the first epitaxial layer penetrates the metal-containing contact layer.
10. The semiconductor device of claim 9, wherein the at least one fin structure includes a first fin

structure and a second fin structure spaced apart from the first fin structure along a second direction crossing the first direction.

11. The semiconductor device of claim 10, wherein the second epitaxial layer connects the first epitaxial layer over the first fin structure and the first epitaxial layer over the second fin structure.

12. The semiconductor device of claim 10, further comprising a void between the first epitaxial layer over the first fin structure, the first epitaxial layer over the second fin structure, the second epitaxial layer, and the insulating layer.

13. The semiconductor device of claim 9, wherein the third epitaxial layer further comprises B in an amount of 5.0×10^{19} atoms/cm³ to 1.0×10^{21} atoms/cm³.

14. The semiconductor device of claim 9, wherein the third epitaxial layer further comprises Ga in an amount of 5.0×10^{19} atoms/cm³ to 1.0×10^{21} atoms/cm³.

15. The semiconductor device of claim 9, wherein the at least one fin structure is made of Si.

16. A semiconductor device comprising: a mesa structure disposed over a substrate; a first fin structure and a second fin structure protruding in a first direction from the mesa structure, wherein the first fin structure and the second fin structure are arranged along a second direction crossing the first direction; an insulating layer disposed over the mesa structure and the first and second fin structures; a first epitaxial layer disposed over first and second fin structures, wherein the first epitaxial layer disposed over the first fin structure is separated from the first epitaxial layer disposed over the second fin structure; a second epitaxial layer disposed over the first epitaxial layer, the second epitaxial layer having a material composition different from the first epitaxial layer and connecting the first fin structure and the second fin structure; a third epitaxial layer disposed over a first portion of the second epitaxial layer, the third epitaxial layer having a material composition different from the second epitaxial layer; and a conductive contact layer disposed over the first epitaxial layer and a second portion of the second layer, wherein the conductive contact layer covers an upper surface and side surfaces of the first epitaxial layer of the first and second fin structures.

17. The semiconductor device of claim 16, wherein the first epitaxial layer comprises Si_{1-x}Ge_x, the second epitaxial layer comprises Si_{1-y}Ge_y, and the third epitaxial layer comprises Si_{1-z}Ge_z, where $x < y$ and $z < y$.

18. The semiconductor device of claim 17, wherein the third epitaxial layer further comprises at least one of B and Ga in an amount of 5.0×10^{19} atoms/cm³ to 1.0×10^{21} atoms/cm³.

19. The semiconductor device of claim 16, wherein a ratio of a height of the first epitaxial layer along the first direction from an upper surface of the insulating layer to a width of the first epitaxial layer along the second direction ranges from 2 to 15.

20. The semiconductor device of claim 16, further comprising a silicide layer and an adhesive layer disposed between the first and second epitaxial layers and the conductive contact layer.
